Evaluating paraphrased answer consistency of large language models in multiple-choice commonsense questions

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Abstract

Despite strong performance on many natural language processing benchmarks, large language models (LLMs) may produce different answers when the same question is expressed using different wording. This raises concerns about the robustness and reliability of LLM evaluation methods that rely on single-prompt benchmarks. This project investigates how consistently LLMs answer multiple-choice commonsense questions when the same question is presented with different paraphrased formulations. Using the CommonsenseQA dataset, we generate paraphrased variants of each question at three difficulty levels light, medium and strong, while preserving the original semantic meaning. Paraphrases are generated using a pretrained T5 paraphrasing model and validated through a BART-large-MNLI natural language inference model, ensuring that generated rewrites preserve the original question's meaning. Three similarly sized open-source LLMs, Llama-2-7B, Mistral-7B, and Yi-6B are evaluated in a zero-shot multiple-choice question answering setting. Model outputs are analysed using accuracy, answer-choice consistency and a paraphrase sensitivity index. Preliminary experiments conducted on a sampled subset of the CommonsenseQA dataset show that all three models experience declining accuracy and consistency as paraphrase strength increases, with Llama-2-7B showing the highest drop in consistency under strong paraphrasing. Semantic filtering generally improves consistency across models. These findings suggest that LLM robustness varies meaningfully across model families and paraphrase intensity, and that consistency is a valuable complement to accuracy as an evaluation metric for commonsense reasoning tasks.

1 Introduction

Large language models (LLMs) have shown strong performance on common sense reasoning tasks, but their answers may still change when the same question is rephrased. Because natural language is sequential data, the same meaning can be expressed through different token sequences, so paraphrasing provides a useful test of whether LLMs rely on semantic understanding or surface wording. This issue is important in current LLM research because small changes in input phrasing can reveal weaknesses in robustness, evaluation reliability, and generalization. Prior work has also shown that benchmark accuracy alone is often insufficient for evaluating NLP systems comprehensively (Ribeiro et al., 2020). However, most existing studies focus on performance differences rather than the consistency of answer choices, especially in multiple-choice common-sense settings. Therefore, this study investigates how consistently several LLMs respond to paraphrased versions of the same common-sense question across different levels of paraphrasing difficulty.

2 Related Work

This section discusses prior research across five major areas directly relevant to our project, LLM robustness and prompt sensitivity, commonsense reasoning benchmarks, evaluation of LLMs on multiple-choice question answering, LLMs being used and paraphrase generation methods.

2.1 LLM Robustness and Prompt Sensitivity

A TACL paper demonstrates that single prompt benchmarks substantially overestimate model capability (Mizrahi et al., 2024). The same model can rank first on one phrasing of a question and last on another. Their call for multi-prompt evaluation directly motivates our use of three paraphrase difficulty levels (weak, medium, strong) instead of a single rewrite condition. Similarly, another finding (Zhuo et al., 2024) introduces ProSA, a framework that operationalizes prompt sensitivity as a proxy for monitoring LLM robustness and stability. They show that dataset level averages can hide instance level sensitivity. Models may seem robust overall but fail on specific questions. This issue is described as paraphrase divergence, where a model answers the original question correctly, but fails on a semantically equivalent rewrite (Fu et al., 2024).

Carranza (Carranza, 2025) finds clear accuracy drops when benchmark questions are rephrased under paraphrase stress tests, confirming that surface-form brittleness is a broad concept. Raj et al. (Raj et al., 2025) did a study on semantic consistency in free form LLM outputs but note that consistency is difficult to quantify without a fixed answer set. Our work fills the gap they identify by focusing on multiple-choice tasks where each question has a finite, labelled option space, enabling precise consistency measurement.

2.2 Commonsense Reasoning Benchmarks

CommonsenseQA (Talmor et al., 2019), presented at NAACL, is a dataset that has become the standard benchmark for evaluating general commonsense reasoning in NLP, with human performance at 89% and strong BERT-based baselines at 56% accuracy. CommonsenseQA was extended with free-text explanations, showing that models often rely on surface cues instead of true conceptual understanding (Aggarwal et al., 2021). Beyond CommonsenseQA, the broader commonsense reasoning literature includes benchmarks such as HellaSwag (Zellers et al., 2019) and Social IQa (Sap et al., 2019), both of which test situational and physical commonsense. The CRoW benchmark measures robustness in commonsense reasoning through changes in overall accuracy (Ismayilzada et al., 2023), whereas our approach evaluates whether model answers remain consistent across different paraphrase strengths.

2.3 Paraphrase Generation Methods

In previous work done for paraphrase generation in the field of Natural Language Processing (NLP) Text-to-Text Transfer Transformer (T5) is a popular algorithm that is commonly used (Widjaja et al., 2022). One paper (Betz and Richardson, 2022) looks at the accuracy and precision of the T5 model and compares it with the Back-Translation guided multi-round Paraphrase Generation (BTmPG) model and found the T5 model outperformed the BTmPG model. This work used the Microsoft Research Paraphrase Corpus (MRPC) and Quora Question Paired (QQP) datasets to pretrain the models and compare the ROUGE1 and BLEU scores. A T5 model was set up and trained with Deep Augmented Analysis (DeepA2) to advance the generated texts given to the model (Betz and Richardson, 2022).

2.4 LLMs being used for the Comparative Study

From a study conducted in 2024 that looked at problematic questions in common-sense datasets, they compared LLMs both from similar model families and different families to see how well they match the gold standard answer (Palta et al., 2024). When looking at selecting models, it is important to select models with similar parameter sizes of different model families to be able to evaluate how well the models perform. The Llama-2-7B, Mistral-7B, and Yi-6B large language models were selected from the study. This was decided because these models are similar in parameter size and were compared with each other in a study that looked at a Social Intelligence QA and CommonsenseQA datasets, so they are known handle multiple choice QA datasets (Palta et al., 2024). Llama-2-7B is an open-source model release from Meta in 2023 that included a collection of pretrained and fine-tuned LLMs that are optimized for dialogue use cases (Touvron et al., 2023). Mistral-7B is an open source model released in 2023 (Jiang et al., 2023). This model leverages grouped-query attention which allows it to accelerate the inference speed and reduced memory requirements. Yi-6B is a release from the Yi model family by 01.AI in 2025, and the finetuned model has a strong human preference rate and performs strongly in a wide range of set benchmarks (Young et al., 2025).

2.5 Evaluation of LLMs on Multiple-Choice Question Answering

Multiple-choice QA is among the most widely used approach for LLM evaluation. Pezeshkpour and Hruschka (Pezeshkpour and Hruschka, 2023), show that LLMs are highly sensitive to the ordering of answer options in MCQs. Some other work demonstrates that LLMs are not reliable multiple-choice selectors, displaying systematic bias toward certain option letters (Zheng et al., 2024),and models are brittle when the correct answer is not among the options (Góral et al., 2025).

Natural Language Inference (NLI) is a task that looks at the concepts of entailment and contradiction for natural languages in computational systems (Bowman et al., 2015). NLI has been essential in use cases that range from semantic parsing to commonsense reasoning. It has the job of determining whether the output makes logical sense or aligns with the original input. The classification falls into three categories, entailment, contradiction, or neutral.

One paper from 2024 does a study on whether multiple choice questions can be useful in detecting the abilities of LLMs, and the results are evaluated for accuracy and consistency, then the relationship between the two metrics are compared in both multiple choice and long-form generation questions (Li et al., 2024). Accuracy is the most used metric for benchmarking purposes and performance assessments. Another proposed method of evaluation was Prompt Sensitivity Index (POSIX), from a study in 2024 that looks to measure the sensitivity of prompts when there are variations (Chatterjee et al., 2024). POSIX was used to evaluate a variety of tasks including classification tasks in the way of multiple-choice questions.

2.6 How this work extends the literature

The existing literature on LLM robustness focuses primarily on accuracy degradation under paraphrasing, without systematically examining whether models choose the same answer across semantically equivalent rewrites. Work on MCQA sensitivity studies option ordering and selection bias, but not question-text variability at controlled difficulty levels. Research on commonsense reasoning benchmarks has not yet produced a paraphrase-augmented version of CommonsenseQA with graded difficulty annotations. Our project addresses all three gaps by: (1) shifting the primary metric from accuracy to answer-choice consistency (2) introducing a three-level paraphrase difficulty taxonomy applied to a commonsense MCQA benchmark and (3) extending the CommonsenseQA dataset with light, medium and strong paraphrase variants for each question.

3 Methodology

3.1 Dataset and Task Definition

We use the CommonsenseQA dataset, a benchmark dataset designed to evaluate commonsense reasoning in natural language processing systems. The dataset consists of approximately 12,000 multiple-choice questions, where each question is associated with five candidate answers (A-E) and one correct answer. The ground-truth labels were created through human annotation and validation, making the dataset suitable for supervised evaluation.

In this project, the task is formulated as a multiple-choice question answering problem. Given a question and multiple options, the model must select the option corresponding to the correct label. While the dataset is typically used to evaluate model accuracy, our work focuses on answer consistency under paraphrasing. To enable this evaluation, we extend the dataset by generating multiple paraphrased versions of each question while preserving its original meaning. By presenting the original and paraphrased questions as a sequence of semantically equivalent inputs, we can measure how stable model predictions remain when the wording of a question changes.

3.2 Data Preprocessing

The CommonsenseQA dataset is already well structured and labelled, therefore only moderate preprocessing is required. After loading the dataset from the Hugging Face library, relevant fields such as the question, choices, and answerkey are extracted and organized into a structured data frame. Some preprocessing steps were needed to clean and normalize the questions, such as formatting inconsistencies and unnecessary whitespace were removed. Second, questions that are too short or unsuitable for paraphrasing are filtered out to ensure meaningful paraphrase generation.

3.3 Model Selection

A small text-to-text T5 model will be used for the initial paraphrase generation at the three levels of reconstruction. This model will reduce computation time and memory costs. Once the paraphrases of the questions have been completed, they will be inputted into three different models of similar parameter sizes to perform a comparative analysis. For the evaluation the Llama-2-7B, Mistral-7B, and Yi-6B models will be used.

3.4 System Pipeline

The system pipeline follows a structured evaluation, as seen in Figure 1, to assess model robustness to paraphrased inputs. The CommonsenseQA dataset is expanded into light, medium, and strong paraphrase variants using the 'T5_Paraphrase_Paws' model that is specifically

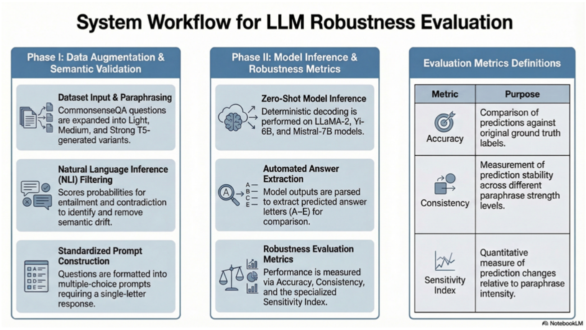

Figure 1: System workflow for LLM robustness evaluation

trained to generate paraphrases, along with filtering applied to remove low-quality outputs. Next, semantic similarity between generated paraphrases and their original question is evaluated using a Natural Language Inference model which computes entailment, neutrality, and contradiction probabilities to determine semantic drift. This step produces separate datasets for unfiltered versus semantically filtered for the following zero-shot model analysis. Each model is set with a multiple-choice prompt containing the question text, answer options, and instructions to respond with a single letter. The prompts are then evaluated using LlaMa2-7B, Yi-6B, and Mistral-Instruct-7B in a zero-shot setting, where outputs are processed and identified by the predicted answer letter from generated responses. Lastly, performance is evaluated using multiple metrics such as accuracy, consistency, and a sensitivity index.

3.5 Compute and Feasibility

All experiments were conducted using Google Colab, a cloud-based computational resource. Original experiment execution used the T4 GPU which did not provide enough memory and runtime resources; thus, the A100 GPU was used to accommodate larger models. Additionally, to reduce GPU memory requirements, models were loaded using 4-bit quantization. The total compute time for all experiments was approximately 15 hours. Individual experiment runs required between 1.5 and 4 hours, depending on the number of paraphrase variants processed.

4 Evaluation Framework

Once the paraphrasing was done by the T5 model, the BART-large-MNLI model was used to compare the similarities and drift within the semantic paraphrases and the original question. The probability of drift in the questions is categorized into entailment, meaning the paraphrased questions are closely related to the original, neutral, or contradiction, meaning the paraphrased question is not semantically similar to the original.

To perform the comparative analysis on the three LLMs being used to answer the paraphrased questions in the dataset, accuracy, consistency and prompt sensitivity will be measured. Accuracy is one of the most common evaluation metrics used for performance assessment, it looks at whether the computed answer matches the ground truth response. Accuracy will be compared at each paraphrasing level for each of the three models. The second metric being used to compare the models is consistency, which determines the degree at which the LLMs provide the same answer when the same question is asked multiple times, for this project each paraphrase level for each question is asked three times. It is not about whether the answer is correct, but rather if how many times the same answer will be provided by the model. The third method used to evaluate the models is a specific method that will determine the model sensitivity, Prompt Sensitivity Index (POSIX) that measured how sensitive the models will be to any changes in the prompt, in this case how changing the paraphrasing levels of questions will affect the answer. For this method prompts must be semantically similar, and POSIX will catch any relative changes of a response when the question is replaced with a different intent-preserving prompt.

5 Experiments and Process

5.1 Experimental Setup

CommonsenseQA dataset, 2000 questions were randomly sampled to form the experimental pool. Paraphrases were generated using the T5 paraphrasing model (Vamsi/T5_Paraphrase_Paws). For each question, paraphrases were produced at three levels of rewriting strength: light, medium, and strong. These levels are controlled through sampling parameters including temperature, top-p, and top-k, which regulate the degree of lexical and structural variation during generation. Specifically, the light level uses temperature = 0.7, top-p = 0.85, and top-k = 80. The medium level uses temperature = 0.95, top-p = 0.90, and top-k = 120. The strong level uses temperature = 1.2, top-p = 0.95, and top-k = 200. For each question, three paraphrases were generated at each level. The generated paraphrases were initially stored in a wide format where each row contains the original question and its associated paraphrase columns. For evaluation, the dataset is transformed into a long format in which each row corresponds to a single question variant. Each entry includes the question text, paraphrase level, answer choices, and ground-truth label.

Three LLMs are evaluated in a zero-shot setting: Llama-2-7B, Mistral-7B, and Yi-6B. To enable efficient inference, all models are loaded using 4-bit quantization with the BitsAndBytes library. All models use the same prompt template:

> Answer the following multiple-choice question. Question: question Choices: A. A B. B C. C D. D E. E. Reply with only one capital letter: A, B, C, D, or E.

This format ensures that all evaluated models receive identical instructions and produce outputs in a consistent format suitable for automatic evaluation. To ensure semantic equivalence between the original question and generated paraphrases, a semantic validation step is applied using the natural language inference model (BART-large-MNLI). For each pair, the original question is treated as the premise, and the paraphrased question as the hypothesis. A paraphrase is accepted if the entailment probability is at least 0.70 and the contradiction probability is at most 0.15.

5.2 Experiment 1: Baseline Analysis

The first experiment evaluates the performance of LLMs to minor paraphrasing. For each question, the evaluation includes the original question and three light paraphrases generated by the paraphrasing model. Two evaluation settings are considered: (1) using all generated paraphrases directly and (2) applying semantic filtering using the NLI validation described in Section 5.1 to retain only paraphrases that satisfy the entailment and contradiction thresholds. For each question variant, the standardized prompt is constructed and evaluated using zero-shot inference. The predicted answer letter (A–E) is extracted from the model output and used to compute accuracy by comparison with the ground-truth label and consistency by comparison with the prediction on the original question.

5.3 Experiment 2: Improved Method

This experiment extends the baseline analysis by evaluating model performance under stronger paraphrasing. For each question, the evaluation includes the original question together with three medium and three strong paraphrases generated by the paraphrasing model. Similar to Experiment 1, two evaluation settings are considered: using all generated paraphrases and applying semantic filtering using the NLI validation described in Section 5.1. Each question variant is evaluated using the standardized prompt in a zero-shot setting, and the predicted answer letter (A–E) is extracted from the model output. Accuracy and consistency are then computed to examine how model predictions change as paraphrase intensity increases and to compare differences across the evaluated LLMs. POSIX will then be looked at to compare how sensitive each of the models are to changes in question prompts. Looking at this metric will be a strong indicator of which model is most sensitive and which are more robust.

5.4 Preliminary Results and Figures

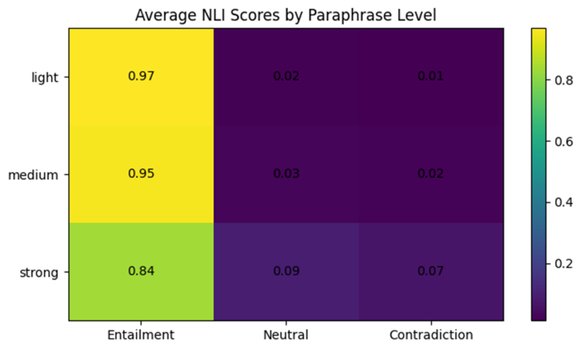

Figure 2: Semantic similarity map of the average NLI scores by each paraphrase level

Figure 2 summarizes the average Natural Language Inference scores between the original questions and their paraphrased versions across three paraphrase levels. For light and medium paraphrases, the entailment scores remain very high (0.97 and 0.95), indicating that most generated paraphrases preserve the original meaning. However, as paraphrasing strength increases, the entailment score drops to 0.84 while neutral and contradiction scores increase. It means that stronger paraphrases more frequently alter the semantic relationship with the original question. This can be seen as a tradeoff between generating diverse paraphrases and maintaining semantic equivalence. A limitation of this method is that the scores are averaged across questions and depend on the NLI model's settings, which may not perfectly reflect true semantic equivalence in all cases.

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